

35. Independent Watchdog Timer (IWDTa)

In this section, “PCLK” is used to refer to PCLKB.

35.1 Overview

The independent watchdog timer (IWDT) can be used to detect programs being out of control.

The user can detect when a program runs out of control if an underflow occurs, by creating a program that refreshes the IWDT counter before it underflows.

The functions of the IWDT are different from those of the WDT in the following respects.

- The divided IWDT-dedicated clock (IWDTCLK) is used as the count source (not affected by the PCLK).
- When making a transition to sleep mode, software standby mode, deep software standby mode, or all-module clock stop mode, the IWDTCSLTPR.SLCSTP bit or the OFS0.IWDTSLCSTP bit can be used to select whether to stop the counter or not.

Table 35.1 lists the specifications of the IWDT and Figure 35.1 shows a block diagram of the IWDT.

Table 35.1 IWDT Specifications

Item	Description
Count source*1	IWDT-dedicated clock (IWDTCLK)
Clock divide ratio	Divide by 1, 16, 32, 64, 128, or 256
Counter operation	Counting down using a 14-bit down-counter
Conditions for starting the counter	• Auto-start mode: Counting automatically starts after a reset is released • Register start mode: Counting is started by refresh operation (writing 00h and then FFh to the IWDTRR register).
Conditions for stopping the counter	• Reset • In low power consumption states (depends on the register setting*2) • A counter underflows or a refresh error occurs (only in register start mode)
Window function	Window start and end positions can be specified (refresh-permitted and refresh-prohibited periods)
Reset output sources	• Down-counter underflows • Refreshing outside the refresh-permitted period (refresh error)
Non-maskable interrupt/ interrupt sources	• Down-counter underflows • Refreshing outside the refresh-permitted period (refresh error)
Reading the counter value	The down-counter value can be read by the IWDTSR register.
Event link function (output)	• Down-counter underflow event output • Refresh error event output
Output signal (internal signal)	• Reset output • Interrupt request output • Sleep mode count stop control output
Auto-start mode (controlled by option function select register 0 (OFS0))	• Selecting the clock frequency divide ratio after a reset (OFS0.IWDTCKS[3:0] bits) • Selecting the timeout period of the independent watchdog timer (OFS0.IWDTTOPS[1:0] bits) • Selecting the window start position in the independent watchdog timer (OFS0.IWDRPSS[1:0] bits) • Selecting the window end position in the independent watchdog timer (OFS0.IWDRPES[1:0] bits) • Selecting the reset output or interrupt request output (OFS0.IWDRSTIRQS bit) • Selecting the down-count stop function at transition to sleep mode, software standby mode, deep software standby mode, or all-module clock stop mode (OFS0.IWDTSLCSTP bit)
Register start mode (controlled by the IWDT registers)	• Selecting the clock frequency divide ratio after refreshing (IWDTCR.CKS[3:0] bits) • Selecting the timeout period of the independent watchdog timer (IWDTCR.TOPS[1:0] bits) • Selecting the window start position in the independent watchdog timer (IWDTCR.RPSS[1:0] bits) • Selecting the window end position in the independent watchdog timer (IWDTCR.RPES[1:0] bits) • Selecting the reset output or interrupt request output (IWDTCCR.RSTIRQS bit) • Selecting the down-count stop function at transition to sleep mode, software standby mode, deep software standby mode, or all-module clock stop mode (IWDTCSLTPR.SLCSTP bit)

Note 1. Satisfy the frequency of the peripheral module clock (PCLK) $\geq 4 \times$ (the frequency of the count source after divide).

Note 2. When the OFS0.IWDTSLCSTP bit is 1 in auto-start mode, and when the IWDTCSLTPR.SLCSTP bit is 1 in register start mode.

To use the IWDT, the IWDT-dedicated clock (IWDTCCLK) should be supplied so that the IWDT operates even if the peripheral module clock (PCLK) stops. The bus interface and registers operate with PCLK, and the 14-bit counter and control circuits operate with IWDTCCLK.

Figure 35.1 is a block diagram of the IWDT.

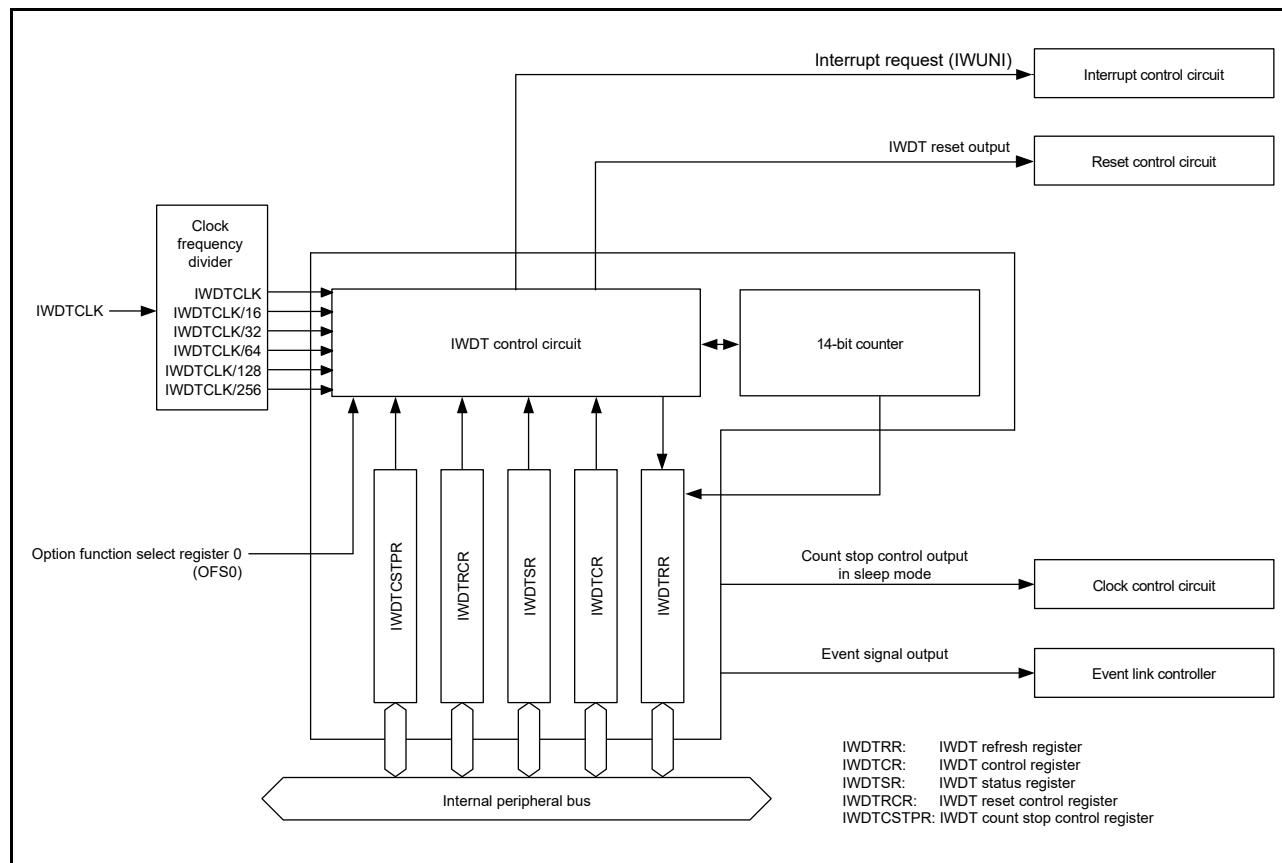
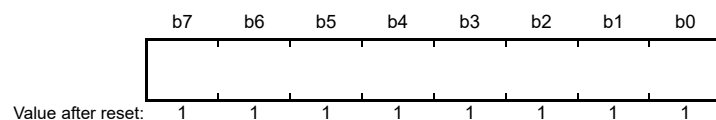


Figure 35.1 IWDT Block Diagram

35.2 Register Descriptions

35.2.1 IWDT Refresh Register (IWDTRR)

Address(es): IWDT.IWDTRR 0008 8030h



Bit	Description	R/W
b7 to b0	The counter is refreshed by writing 00h and then writing FFh to this register.	R/W

The IWDTRR register refreshes the counter of the IWDT.

The counter of the IWDT is refreshed by writing 00h and then writing FFh to the IWDTRR register (refresh operation) within the refresh-permitted period.

After the counter has been refreshed, it starts counting down from the value selected by the IWDTTOPS[1:0] bits in option function select register 0 (OFS0) in auto-start mode. In register start mode, counting down starts from the value selected by setting the IWDTCR.TOPS[1:0] bits in the first refresh operation after a reset is released.

When 00h is written, the read value is 00h. When a value other than 00h is written, the read value is FFh.

For details of the refresh operation, refer to [section 35.3.3, Refresh Operation](#).

35.2.2 IWDt Control Register (IWDTCR)

Address(es): IWDt.IWDTCR 0008 8032h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	RPSS[1:0]	—	—	RPES[1:0]			CKS[3:0]				—	—	TOPS[1:0]	
Value after reset:	0	0	1	1	0	0	1	1	1	1	1	1	0	0	1	1

Bit	Symbol	Bit Name	Description	R/W
b1, b0	TOPS[1:0]	Timeout Period Select	b1 b0 0 0: 1024 cycles (03FFh) 0 1: 4096 cycles (0FFFh) 1 0: 8192 cycles (1FFFh) 1 1: 16384 cycles (3FFFh)	R/W
b3, b2	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b7 to b4	CKS[3:0]	Clock Divide Ratio Select	b7 b4 0 0 0 0: No division 0 0 1 0: Divide-by-16 0 0 1 1: Divide-by-32 0 1 0 0: Divide-by-64 1 1 1 1: Divide-by-128 0 1 0 1: Divide-by-256 Other settings are prohibited.	R/W
b9, b8	RPES[1:0]	Window End Position Select	b9 b8 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (window end position is not specified.)	R/W
b11, b10	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b13, b12	RPSS[1:0]	Window Start Position Select	b13 b12 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (window start position is not specified.)	R/W
b15, b14	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R

There are some restrictions on writing to the IWDTCR register. For details, refer to section 35.3.2, Control over Writing to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers.

In auto-start mode, the settings in the IWDTCR register are disabled, and the settings in option function select register 0 (OFS0) are enabled. The bit setting made to the IWDTCR register can also be made in the OFS0 register. For details, refer to section 35.3.8, Correspondence between Option Function Select Register 0 (OFS0) and IWDt Registers.

TOPS[1:0] Bits (Timeout Period Select)

These bits select the timeout period (period until the counter underflows) from among 1024, 4096, 8196, or 16384 cycles, taking the divided clock specified by the CKS[3:0] bits as one cycle.

After the counter is refreshed, the combination of the CKS[3:0] and TOPS[1:0] bits determines the time (number of IWDTCCLK cycles) until the counter underflows.

Relations between the CKS[3:0] and TOPS[1:0] bit setting, the timeout period, and the number of IWDTCCLK cycles are listed in Table 35.2.

Table 35.2 Settings and Timeout Periods

CKS[3:0] Bits				TOPS[1:0] Bits		Clock Divide Ratio	Timeout Period (Number of Cycles)	Cycles of IWDTCCLK
b7	b6	b5	b4	b1	b0			
0	0	0	0	0	0	No division	1024	1024
				0	1		4096	4096
				1	0		8192	8192
				1	1		16384	16384
0	0	1	0	0	0	Divide-by-16	1024	16384
				0	1		4096	65536
				1	0		8192	131072
				1	1		16384	262144
0	0	1	1	0	0	Divide-by-32	1024	32768
				0	1		4096	131072
				1	0		8192	262144
				1	1		16384	524288
0	1	0	0	0	0	Divide-by-64	1024	65536
				0	1		4096	262144
				1	0		8192	524288
				1	1		16384	1048576
1	1	1	1	0	0	Divide-by-128	1024	131072
				0	1		4096	524288
				1	0		8192	1048576
				1	1		16384	2097152
0	1	0	1	0	0	Divide-by-256	1024	262144
				0	1		4096	1048576
				1	0		8192	2097152
				1	1		16384	4194304

CKS[3:0] Bits (Clock Divide Ratio Select)

These bits select the IWDTCCLK clock divide ratio from among divide-by 1, 16, 32, 64, 128, and 256. Combination with the TOPS[1:0] bit setting, a count period between 1024 and 4194304 cycles of the IWDTCCLK clock can be selected for the IWDTC.

RPES[1:0] Bits (Window End Position Select)

These bits select 75%, 50%, 25% or 0% of the count period for the window end position of the counter. The window end position should be a value smaller than the window start position (window start position > window end position). If the window end position is greater than the window start position, only the window start position setting is enabled.

The counter values for the window start and end positions selected by setting the RPSS[1:0] and RPES[1:0] bits change depending on the TOPS[1:0] bit setting.

Table 35.3 lists the counter values for the window start and end positions corresponding to TOPS[1:0] bit values.

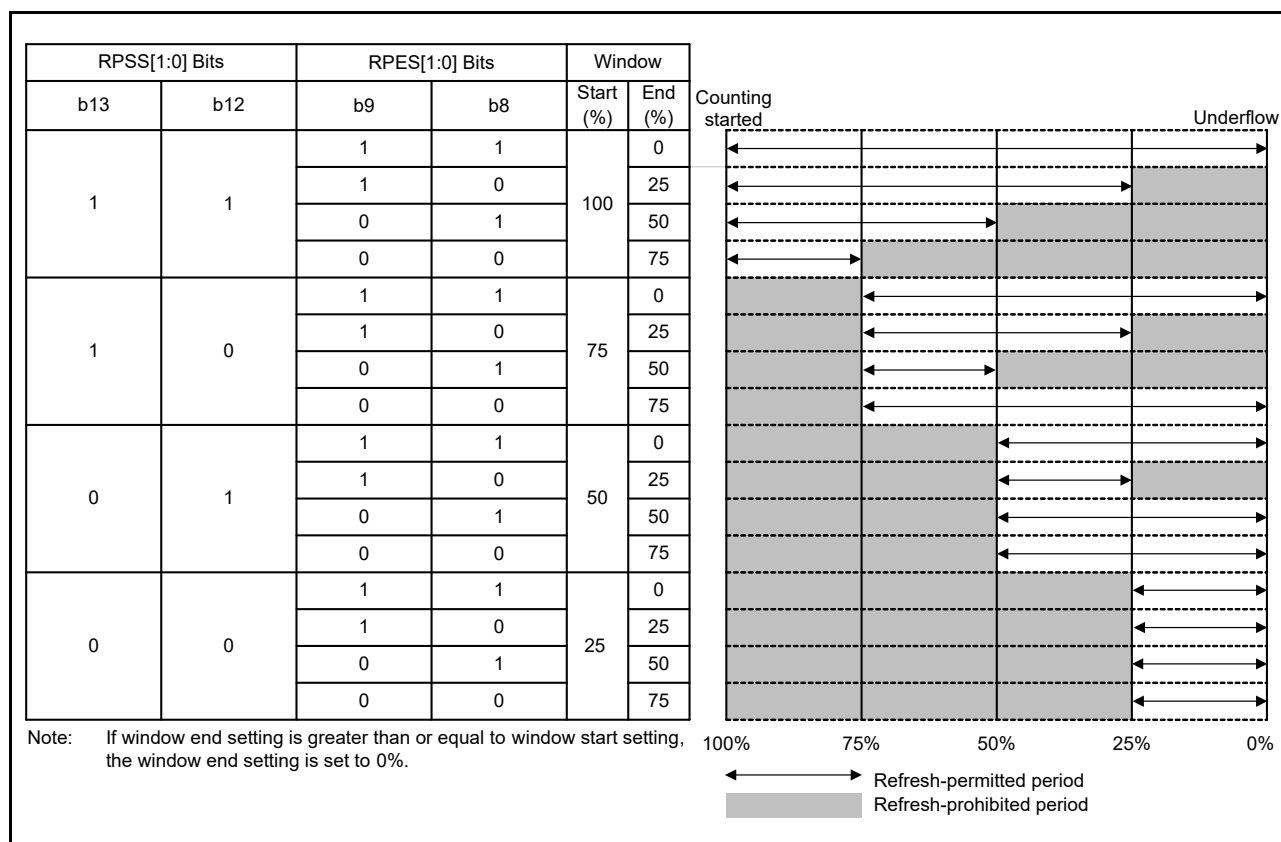
Table 35.3 Relationship between Timeout Period and Window Start and End Counter Values

TOPS[1:0] Bits		Timeout Period		Window Start and End Counter Value			
b1	b0	Cycles	Counter Value	100%	75%	50%	25%
0	0	1024	03FFh	03FFh	02FFh	01FFh	00FFh
0	1	4096	0FFFh	0FFFh	0BFFh	07FFh	03FFh
1	0	8192	1FFFh	1FFFh	17FFh	0FFFh	07FFh
1	1	16384	3FFFh	3FFFh	2FFFh	1FFFh	0FFFh

RPSS[1:0] Bits (Window Start Position Select)

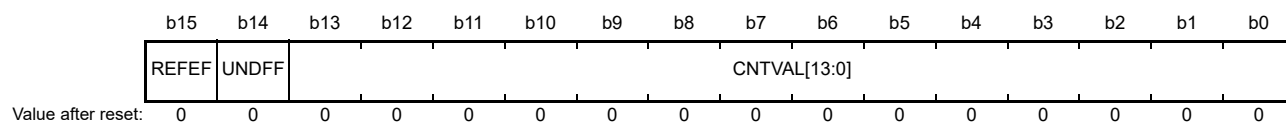
These bits select a counter window start position from 100%, 75%, 50%, or 25% of the count period (100% when the count starts and 0% when the counter underflows). The interval between the window start position and window end position is the refresh-permitted period and the other periods are refresh-prohibited periods.

Figure 35.2 shows the relationship between of the RPSS[1:0] and RPES[1:0] bit setting and the refresh-permitted and refresh-prohibited periods.

**Figure 35.2 RPSS[1:0] and RPES[1:0] Bit Settings and the Refresh-Permitted Period**

35.2.3 IWDt Status Register (IWDTSR)

Address(es): IWDt.IWDTSR 0008 8034h



Bit	Symbol	Bit Name	Description	R/W
b13 to b0	CNTVAL[13:0]	Counter Value	Value counted by the counter	R
b14	UNDF	Underflow Flag	0: No underflow occurred 1: Underflow occurred	R/(W) *1
b15	REFEF	Refresh Error Flag	0: No refresh error occurred 1: Refresh error occurred	R/(W) *1

Note 1. Only 0 can be written to clear the flag.

CNTVAL[13:0] Bits (Counter Value)

These bits are used to confirm the counter value of the counter, but note that the read value may differ from the actual count by a value of one count.

UNDF Flag (Underflow Flag)

This bit is used to confirm whether or not an underflow has occurred in the counter.

The value 1 indicates that the counter has underflowed. The value 0 indicates that the counter has not underflowed.

Write 0 to the UNDF flag to set the value to 0. Writing 1 has no effect.

REFEF Flag (Refresh Error Flag)

This bit is used to confirm whether or not a refresh error (performing a refresh operation during a refresh-prohibited period).

The value 1 indicates that a refresh error has occurred. The value 0 indicates that no refresh error has occurred.

Write 0 to the REFEF flag to set the value to 0. Writing 1 has no effect.

35.2.4 IWDt Reset Control Register (IWDTRCR)

Address(es): IWDt.IWDTRCR 0008 8036h

	b7	b6	b5	b4	b3	b2	b1	b0
	RSTIR QS	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b7	RSTIRQS	Reset Interrupt Request Select	0: Non-maskable interrupt request or interrupt request output is enabled. 1: Reset output is enabled.	R/W

There are some restrictions on writing to the IWDTRCR register. For details, refer to section 35.3.2, Control over Writing to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers.

In auto-start mode, the IWDTRCR register setting are disabled, and the settings in option function select register 0 (OFS0) are enabled. The bit setting mode to the IWDTRCR register can also be made in the OFS0 register. For details, refer to section 35.3.8, Correspondence between Option Function Select Register 0 (OFS0) and IWDt Registers.

35.2.5 IWDt Count Stop Control Register (IWDTCSTPR)

Address(es): IWDt.IWDTCSTPR 0008 8038h

	b7	b6	b5	b4	b3	b2	b1	b0
	SLCST P	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b7	SLCSTP	Sleep Mode Count Stop Control	0: Count stop is disabled. 1: Count is stopped at a transition to sleep mode, software standby mode, deep software standby mode, or all-module clock stop mode.	R/W

The IWDTCSTPR register controls whether to stop the IWDt counter in a low power consumption state. There are some restrictions on writing to the IWDTCSTPR register. For details, refer to section 35.3.2, Control over Writing to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers.

In auto-start mode, the IWDTCSTPR register setting are disabled, and the settings in option function select register 0 (OFS0) are enabled. The bit setting mode to the IWDTCSTPR register can also be made in the OFS0 register. For details, refer to section 35.3.8, Correspondence between Option Function Select Register 0 (OFS0) and IWDt Registers.

SLCSTP Bit (Sleep Mode Count Stop Control)

This bit selects whether to stop counting at a transition to sleep mode, software standby mode, deep software standby mode, or all-module clock stop mode.

35.2.6 Option Function Select Register 0 (OFS0)

For option function select register 0 (OFS0), refer to section 35.3.8, Correspondence between Option Function Select Register 0 (OFS0) and IWDt Registers.

35.3 Operation

35.3.1 Count Operation in Each Start Mode

Select the IWDt start mode by setting the IWDtSTRT bit in option function select register 0 (OFS0).

When the OFS0.IWDtSTRT bit is 1 (register start mode), the IWDtCR, IWDtRCR, and IWDtCSTPR registers are enabled, and counting is started by refresh operation (writing) to the IWDtRR register. When the OFS0.IWDtSTRT bit is 0 (auto-start mode), the setting of the OFS0 register is enabled, and counting automatically starts after reset.

35.3.1.1 Register Start Mode

When the OFS0.IWDtSTRT bit in option function select register 0 is 1, register start mode is selected, and the IWDtCR, IWDtRCR, and IWDtCSTPR registers are enabled.

After a reset is released, set the clock divide ratio, window start and end positions, and timeout period in the IWDtCR register, the reset output or interrupt request output in the IWDtRCR register, and the counter stop control at transitions to low power consumption states in the IWDtCSTPR register. Then refresh the counter to start counting down from the value selected by setting the IWDtCR.TOPS[1:0] bits.

Thereafter, as long as the program continues normal operation and the counter is refreshed in the refresh-permitted period, the value in the counter is re-set each time the counter is refreshed and counting down continues. The IWDt does not output the reset signal as long as this continues. However, if the counter underflows because the counter cannot be refreshed due to a program runaway, or if a refresh error occurs because the counter was refreshed outside the refresh-permitted period, the IWDt outputs a reset signal or a non-maskable interrupt request/interrupt request (IWUNI). Set the IWDtRCR.RSTIRQS bit to select either reset output or interrupt request output.

Figure 35.3 shows an example of operation under the following conditions.

- Register start mode (OFS0.IWDtSTRT = 1)
- Reset output is enabled (IWDtRCR.RSTIRQS = 1)
- The window start position is 75% (IWDtCR.RPSS[1:0] = 10b)
- The window end position is 25% (IWDtCR.RPES[1:0] = 10b)

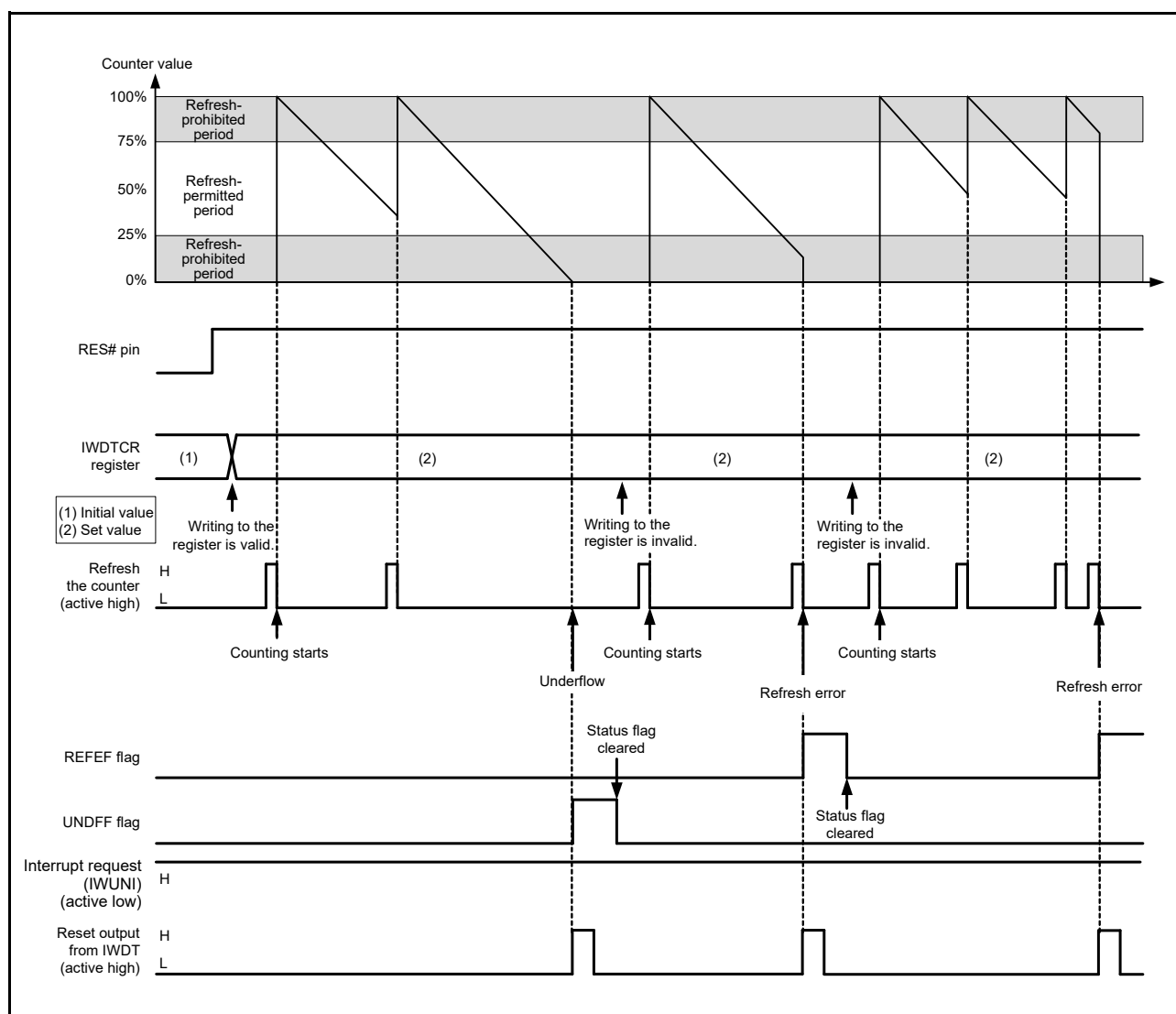


Figure 35.3 Operation Example in Register Start Mode

35.3.1.2 Auto-Start Mode

When the IWDTSTRT bit in option function select register 0 (OFS0) is 0, auto-start mode is selected, and the IWDTCR, IWDTRCR, and IWDTCSTPR registers are disabled.

Within the reset state, the clock divide ratio, window start and end positions, timeout period, reset output or interrupt request output, and counter stop control at transitions to low power consumption states are set using the values specified in the OFS0 register. When the reset is released, the counter automatically starts counting down from the value selected by the OFS0.IWDTTOPS[1:0] bits.

After that, as long as the program continues normal operation and the counter is refreshed in the refresh-permitted period, the value in the counter is re-set each time the counter is refreshed and counting down continues. The IWDt does not output the reset signal as long as this continues. However, if the counter underflows because refreshing of the counter is not possible due to the program having entered crashed execution or if a refresh error occurs due to refreshing outside the refresh-permitted period, the IWDt outputs the reset signal or non-maskable interrupt request/interrupt request (IWUNI). After the reset signal or non-maskable interrupt request/interrupt request (IWUNI) is generated, the counter reloads the timeout period after counting for one cycle, and restarts counting. Set the OFS0.IWDTRSTIRQS bit to select either reset output or interrupt request output.

Figure 35.4 shows an example of operation under the following conditions.

- Auto-start mode (OFS0.IWDTSTRT = 0)
- Non-maskable interrupt request output is enabled (OFS0.IWDTRSTIRQS = 0)
- The window start position is 75% (OFS0.IWDTRPSS[1:0] = 10b)
- The window end position is 25% (OFS0.IWDTRPES[1:0] = 10b)

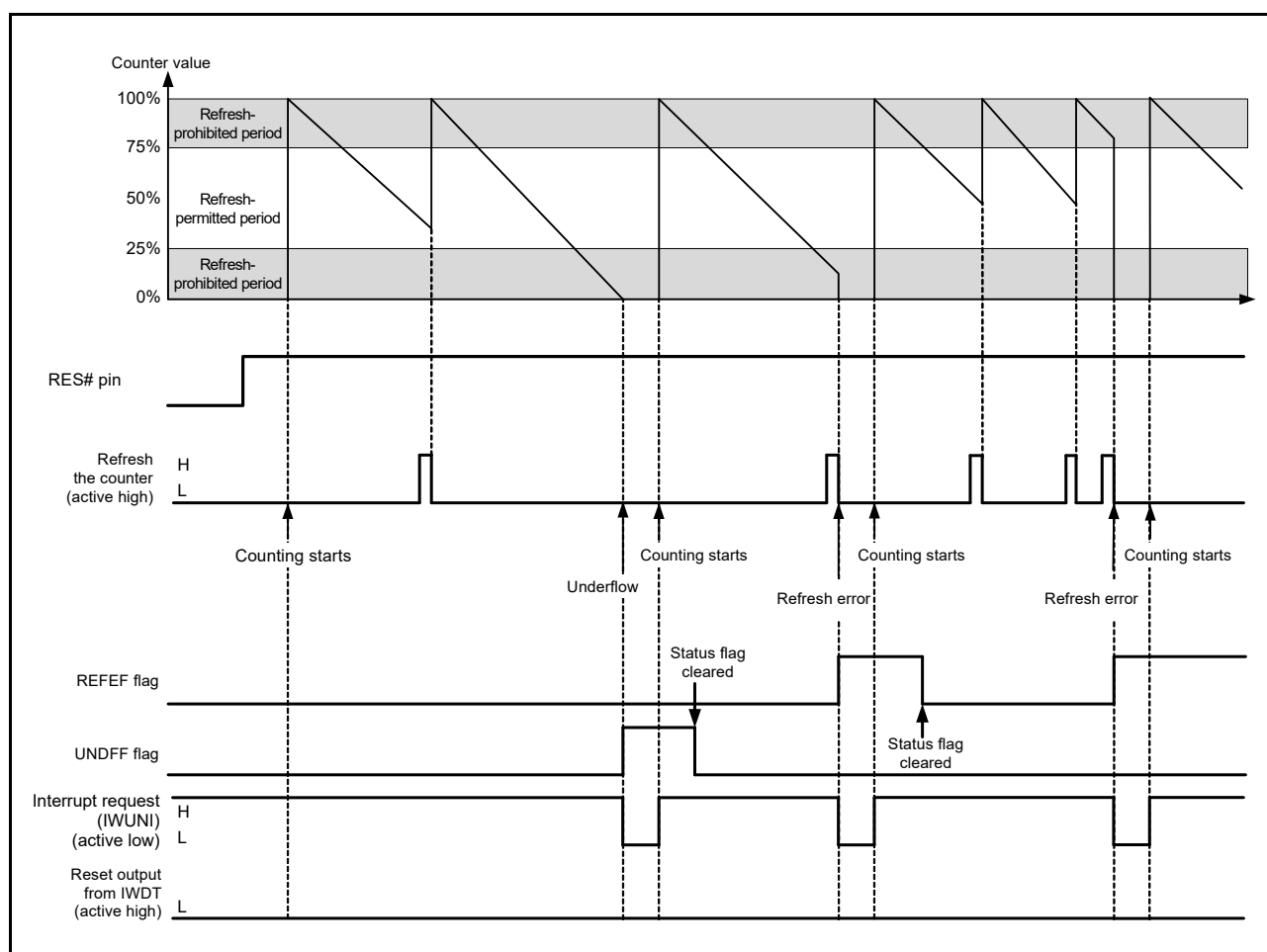


Figure 35.4 Operation Example in Auto-Start Mode

35.3.2 Control over Writing to the IWDTCR, IWDTRCR, and IWDTCSTPR Registers

Writing to the IWDTCR, IWDTRCR, or IWDTCSTPR register is only possible once between the release from the reset state and the first refresh operation.

After a refresh operation (counting starts) or the IWDTCR, IWDTRCR, or IWDTCSTPR register is written to, the protection signal in the IWDt becomes 1 to protect registers IWDTCR, IWDTRCR, and IWDTCSTPR against subsequent attempts at writing.

This protection is released by the reset source of the IWDt. With other reset sources, the protection is not released.

Figure 35.5 shows control waveforms produced in response to writing to the IWDTCR register.

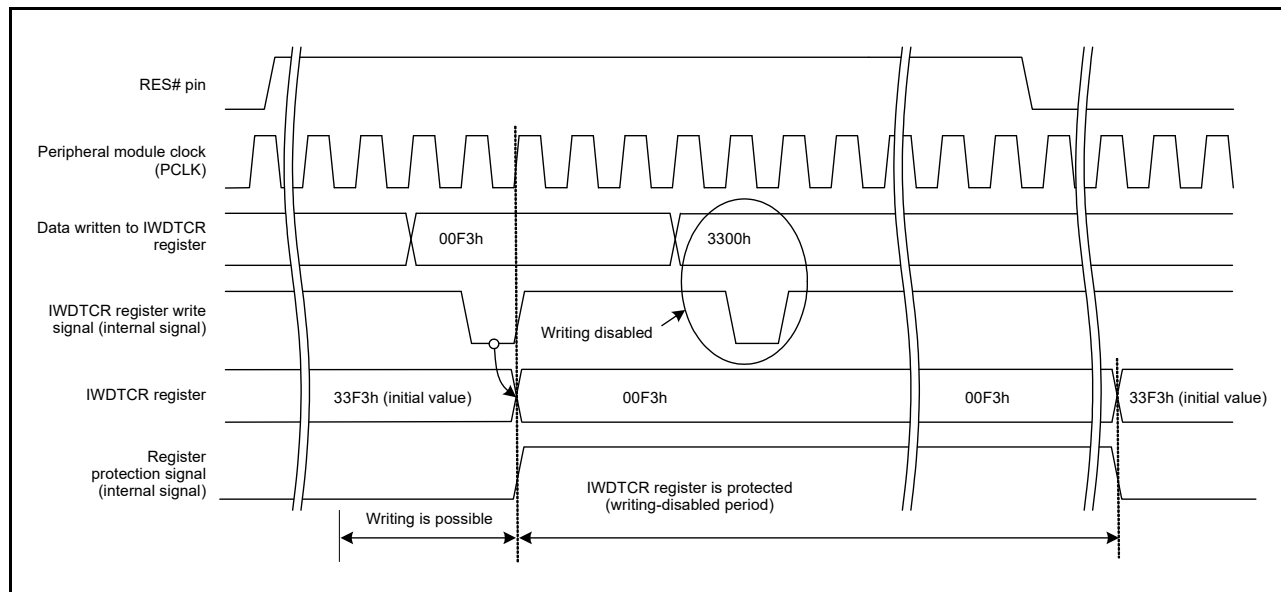


Figure 35.5 Control Waveforms Produced in Response to Writing to the IWDTCR Register

35.3.3 Refresh Operation

The counter is refreshed and starts operation (counting is started by refreshing) by writing the values 00h and then FFh to the IWDTRR register. If a value other than FFh is written after 00h, the counter is not refreshed. After such invalid writing, correct refreshing is performed by again writing 00h and then FFh to the IWDTRR register.

When writing is done in the order of 00h (first time) → 00h (second time), and if FFh is written after that, the writing order 00h → FFh is satisfied; writing 00h (n-1-th time) → 00h (nth time) → FFh is valid and correct refreshing will be done. Even when the first value written before 00h is not 00h, correct refreshing will be done if the operation contains the set of writing 00h → FFh. Moreover, even if a register other than the IWDTRR register is accessed or the IWDTRR register is read between writing 00h and writing FFh to the IWDTRR register, correct refreshing will be done.

[Sample sequences of writing that are valid for refreshing the counter]

- 00h → FFh
- 00h (n-1-th time) → 00h (nth time) → FFh
- 00h → access to another register or read from the IWDTRR register → FFh

[Sample sequences of writing that are not valid for refreshing the counter]

- 23h (a value other than 00h) → FFh
- 00h → 54h (a value other than FFh)
- 00h → AAh (00h and a value other than FFh) → FFh

Even when 00h is written to the IWDTRR register outside the refresh-permitted period, if FFh is written to the IWDTRR register in the refresh-permitted period, the writing sequence is valid and refreshing will be done.

After FFh is written to the IWDTRR register, refreshing the counter requires up to four cycles of the signal for counting (the IWDTCR.CKS[3:0] bits determine how many cycles of the IWDt-dedicated clock (IWDtCLK) make up one cycle for counting). Therefore, writing FFh to the IWDTRR register should be completed four-count cycles before the end position of the refresh-permitted period or a counter underflow. The value of the counter can be checked by the IWDTSR.CNTVAL[13:0] bits.

[Sample refreshing timings]

- When the window start position is set to 1FFFh, even if 00h is written to the IWDTRR register before 1FFFh is reached (2002h, for example), refreshing is done if FFh is written to the IWDTRR register after the value of the IWDTSR.CNTVAL[13:0] bits has reached 1FFFh.
- When the window end position is set to 1FFFh, refreshing is done if 2003h (four-count cycles before 1FFFh) or a greater value is read from the IWDTSR.CNTVAL[13:0] bits immediately after writing 00h → FFh to the IWDTRR register.
- When the refresh-permitted period continues until count 0000h, refreshing can be done immediately before an underflow. In this case, if 0003h (four-count cycles before an underflow) or a greater value is read from the IWDTSR.CNTVAL[13:0] bits immediately after writing 00h → FFh to the IWDTRR register, no underflow occurs and refreshing is done.

Figure 35.6 shows the IWDt refresh-operation waveforms when $PCLK > IWDtCLK$ and clock divide ratio = $IWDtCLK$.

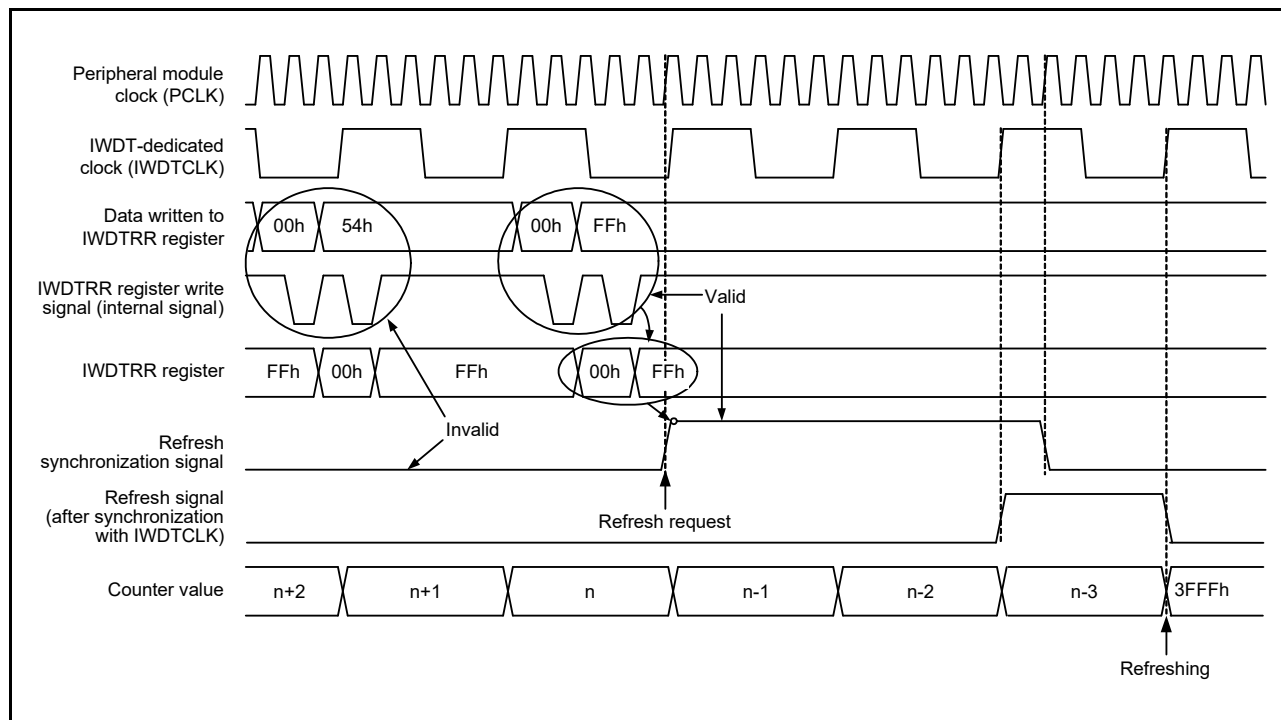


Figure 35.6 IWDt Refresh Operation Waveforms (IWDtCR.CKS[3:0] = 0000b, IWDtCR.TOPS[1:0] = 11b)

35.3.4 Status Flags

The IWDTSR.REFEF and IWDTSR.UNDF flags retain the source of the reset signal output from the IWDt or the source of the interrupt request from the IWDt.

Thus, after release from the reset state or interrupt request generation, read the IWDTSR.REFEF and IWDTSR.UNDF flags to check for the reset or interrupt source.

For each flag, writing 0 clears the bit and writing 1 has no effect.

Leaving the status flags unchanged does not affect operation. If the flags are not cleared, at the time of the next reset or interrupt request from the IWDt, the earlier reset or interrupt source is cleared and the new reset or interrupt source is written.

After 0 is written to each flag, up to three IWDtCLK cycles and two PCLK cycles are required before the value is reflected.

35.3.5 Reset Output

When the IWDTRCR.RSTIRQS bit is set to 1 in register start mode or when the IWDTRSTIRQS bit in option function select register 0 (OFS0) is set to 1 in auto-start mode, a reset signal is output when an underflow in the counter or a refresh error occurs.

In register start mode, the counter is initialized (0000h) and kept in that state after assertion of the reset signal. After the reset is released and the program is restarted, the counter is set up again and counting down is started by refreshing.

In auto-start mode, counting down automatically starts after the reset output.

35.3.6 Interrupt Sources

When the IWDTRCR.RSTIRQS bit is set to 0 in register start mode or when the OFS0.IWDTRSTIRQS bit is set to 0 in auto-start mode, an interrupt (IWUNI) signal is output when an underflow in the counter or a refresh error occurs. This interrupt can be used as a non-maskable interrupt or a maskable interrupt. For details, refer to [section 15, Interrupt Controller \(ICUD\)](#).

Table 35.4 IWDt Interrupt Source

Name	Interrupt Source	DTC Activation	DMAC Activation
IWUNI	Counter underflow Refresh error	Not possible	Not possible

35.3.7 Reading the Counter Value

As the counter in IWDt-dedicated clock (IWDtCLK), the counter value cannot be read directly. The IWDt synchronizes the counter value with the peripheral module clock (PCLK) and stores it in the IWDtSR.CNTVAL[13:0] bits. Thus, the counter value can be checked indirectly through the IWDtSR.CNTVAL[13:0] bits.

Reading the counter value requires multiple PCLK clock cycles (up to four clock cycles), and the read counter value may differ from the actual counter value by a value of one count.

Figure 35.7 shows the processing for reading the IWDt counter value when $PCLK > IWDtCLK$ and clock divide ratio = IWDtCLK.

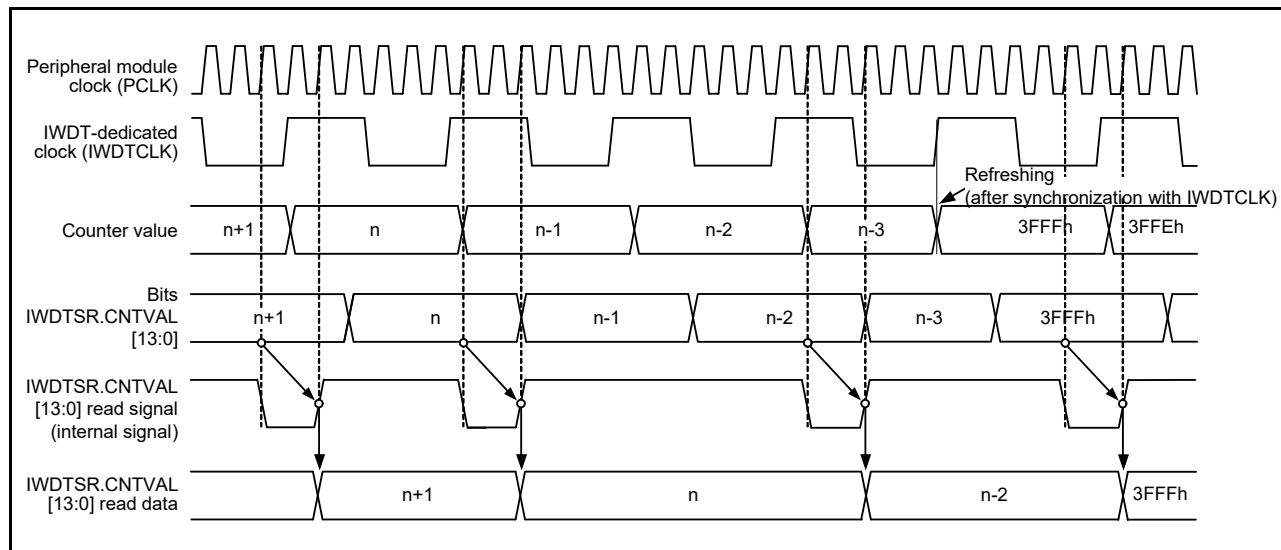


Figure 35.7 Processing for Reading IWDt Counter Value
(IWDtCR.CKS[3:0] = 0000b, IWDtCR.TOPS[1:0] = 11b)

35.3.8 Correspondence between Option Function Select Register 0 (OFS0) and IWDT Registers

Table 35.5 lists the correspondence between option function select register 0 (OFS0) used in auto-start mode and the registers used in register start mode.

Do not change the OFS0 register setting during IWDT operation.

For details on the OFS0 register, refer to section 7.2.3, Option Function Select Register 0 (OFS0).

Table 35.5 Correspondence between Option Function Select Register 0 (OFS0) and IWDT Registers

Target of Control	Function	OFS0 Register (Enabled in Auto-Start Mode) OFS0.IWDTSTRT = 0	IWDT Registers (Enabled in Register Start Mode) OFS0.IWDTSTRT = 1
Counter	Timeout period selection	OFS0.IWDTTOPS[1:0]	IWDTCR.TOPS[1:0]
	Clock frequency divide ratio selection	OFS0.IWDTCKS[3:0]	IWDTCR.CKS[3:0]
	Window start position selection	OFS0.IWDRPSS[1:0]	IWDTCR.RPSS[1:0]
	Window end position selection	OFS0.IWDRPES[1:0]	IWDTCR.RPES[1:0]
Reset output or interrupt request output	Reset output or interrupt request output selection	OFS0.IWDRSTIRQS	IWDTCCR.RSTIRQS
Count stop	Sleep mode count stop control	OFS0.IWDTSLCSTP	IWDTCTPR.SLCSTP

35.4 Link Operation by ELC

The event link controller (ELC) can use the interrupt request signal generated by the IWDT as the event signal.

Therefore, the ELC generates an event to the module specified previously when the IWDT outputs an interrupt request.

The event signal is output by the counter underflow and refresh error.

An event signal is output regardless of the setting of the IWDTCCR.RSTIRQS bit in register start mode or the OFS0.IWDRSTIRQS bit in auto-start mode. An event signal can also be output upon generation of the next interrupt source while the IWDTSR.REFEF or IWDTSR.UNDF flag is 1.

For details, see section 21, Event Link Controller (ELC).

35.5 Usage Notes

35.5.1 Refresh Operations

When making the settings to control the timing of refreshing, consider variations in the range of errors due to the accuracy of the PCLK and IWDTCLK and set values which ensure that refreshing is possible.

35.5.2 Clock Divide Ratio Setting

Satisfy the frequency of the peripheral module clock ($PCLK \geq 4 \times$ (the frequency of the count source after divide)).